

# Heart Disease Severity Prediction & Model Comparison

*A Pure Object-Oriented 11-Step Machine Learning Pipeline with Resampling & Hyperparameter Optimization*

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## 1. Executive Summary & Problem Formulation

Coronary artery disease is a leading global health concern. This project implements an enterprise-grade, 11-step Object-Oriented Machine Learning workflow on the multicenter UCI Heart Disease dataset (920 patients). The pipeline encompasses structured data preparation, visual EDA, statistical transformation, feature selection, stratified partitioning, baseline training, SMOTE resampling, cross-validation, hyperparameter tuning, and comprehensive model comparison.

## 2. Object-Oriented Programming (OOP) Pipeline Architecture

The codebase is organized into 11 dedicated, single-responsibility Python classes coordinated by HeartDiseasePipeline in Main.py:

- **Step 1 (DataPreparation):** Ingests raw data, handles missingness, and corrects 172 zero-cholesterol anomalies.
- **Step 2 (ExploratoryDataAnalysis):** Exports summary statistics and 9 publication-grade figures into EDA/.
- **Step 3 (DataTransformation):** Imputation (median/mode), One-Hot/Binary encoding, and StandardScaler scaling.
- **Step 4 (FeatureSelector):** Prunes non-predictive id and high-missing columns (ca, thal, slope).
- **Step 5 (DataSplitter):** Stratified 80/20 train/test partitioning.
- **Step 6 & 7 (ModelTrainer & ModelEvaluator):** Fits and evaluates Logistic Regression, Random Forest, and XGBoost.
- **Step 8 (DataSampler):** Applies SMOTE resampling to address severe minority class scarcity.
- **Step 9 (CrossValidator):** 5-Fold Stratified Cross-Validation across candidate models.
- **Step 10 (HyperparameterOptimizer):** GridSearchCV hyperparameter tuning on XGBoost.
- **Step 11 (ModelComparator):** Final benchmarking, comparative plots, and serialization of best\_model.pkl.

### 3. Experimental Model Comparison & Benchmark Results

All candidate models were evaluated on the held-out test set (184 unseen samples):

| Model Architecture             | Training Strategy | Test Accuracy | Macro F1 | Weighted F1 |
|--------------------------------|-------------------|---------------|----------|-------------|
| Logistic Regression (Baseline) | Original Split    | 60.87%        | 0.3662   | 0.5801      |
| Random Forest (Baseline)       | Original Split    | 57.07%        | 0.2752   | 0.5077      |
| XGBoost (Baseline)             | Original Split    | 60.33%        | 0.3855   | 0.5859      |
| XGBoost (Tuned + SMOTE)        | SMOTE Resampled   | 56.52%        | 0.4173   | 0.5787      |

### 4. Key Takeaways & Conclusion

- **Impact of SMOTE:** Applying SMOTE improved minority class recall and increased **Macro F1-Score from 0.3855 to 0.4173** on XGBoost, showing improved sensitivity across rare severity stages.
- **Best Model Selection:** Logistic Regression and XGBoost achieved top overall test accuracy (~60.87%), while tuned XGBoost demonstrated superior balanced class discrimination.
- **Delivery Structure:** The repository is equipped with a master Main.py, 11 standalone OOP step classes, dedicated artifact directories (Datapreparation/, EDA/, TransformedData/, Models/, ModelComparison/), an interactive notebook HeartDiseasePrediction.ipynb, and serialized model artifacts.